(b) Explain the working of balanced duplexer with the help of a neat sketch in both transmit and receive mode. 7

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(c) Discuss block diagram of super heterodyne receiver. 7

(d) Discuss about the role of magnetron in radar. 7

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B. E. (Eighth Semester) Examination, April-May, 2016

(Old Scheme)

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(ET & T Engg. Branch)

RADAR ENGINEERING & NAVIGATIONAL AIDS

(Elective-III)

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : *All questions are compulsory with internal choice. Part (a) is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question. Answer should be strcitly to the point.*

Unit-I

1. (a) Define Radar Cross-Section of Target. 2

(b) Derive the Radar Range Equation and mention its significance. 7

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(c) The receiver of a Radar has a Noise Figure of 6 dB. If the 9F banwidh B of the reciever be 3 MHz, the radar is designed for operation at 10 GHz with an antenna of diameter 2 m, calculate the minimum detectable power, and the peak pulse power required to have a maximum range of 1000 km with the target of cross sectional area 20 m? (Assume $T = 290$ K). 7

(d) Derive an expression to find the probability of detection of false alarm in case of radar. Also discuss the false alarm time. 7

Unit-II

2. (a) Explain the function of COHO and STALO in case of MTJ radar. 2

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(b) Explain delay line canceller. Derive an expression for output response of a MTI receiver with delay line canceller. 7

(c) Discuss the block diagram of range gated doppler filter. 7

(d) Describe about Low-Angle Tracking and also discuss the methods for reducing Multipath effects at low Angles. 7

Unit-III

3. (a) Define Standard Propagation. 2

(b) Describe scattrering of radar signal from the round earth's surface. 7

(c) Describe the attenuation effects on the radar signal by atmospheric gases. 7

(d) Explain phenomenon of diffraction in case of propagation of radar waves. 7

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Unit-IV

4. (a) Define Radome. 2

(b) Expalin the working of cassegrain antenna with the help of suitable diagram. 7

(c) Explain about cosecant squared antenna pattern. 7

(d) Explain phased array antenna along with its radiation pattern. 7

Unit-V

5. (a) Define Noise-Figure. 2

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